

Wireless Communication Numerical Companion

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 22 papers, 2070–2082 BS

Ch 2 · Cellular Concepts Numerical Problems & Solutions

Every problem below is a real PYQ, with its year tags and marks. Solutions are worked in full: what was calculated, why, and which formula was used.

Nothing is skipped to save space.

The five formulas everything here is built on

- **Cluster size** $N = i^2 + ij + j^2$ (allowed: 1, 3, 4, 7, 9, 12, 13, 16, 19, 21)
- **Co-channel reuse ratio** $Q = D/R = \sqrt{3N}$
- **Worst-case SIR** $\frac{S}{I} = \frac{(\sqrt{3N})^n}{i_0}$, so $N = \frac{1}{3} \left(i_0 \cdot \frac{S}{I} \right)^{2/n}$
- **Hexagonal cell area** $A = \frac{3\sqrt{3}}{2} R^2 = 2.598 R^2$
- **Traffic** $A_u = \lambda H$, $A = U A_u$, and Erlang B for the GoS

i_0 = number of first-tier interferers: **6** omni, **2** for 120° sectoring, **1** for 60° sectoring. n = path loss exponent.

Problem 1 Prove $Q = \sqrt{3N}$ and find the optimal N

TOP 6/22 71 Bh **4+4**, 79 Bh **3+5**, 80 Bh **4**, 70 Bh **3+4+3**, 81 Bh **3+3**, 74 Ma **4**

1.1 SIR = 15 dB worst case, $n = 4$, considering trunking efficiency (71 Bh)

QUESTION AS PRINTED 2071 Bhadra · Q2 · [4+4]

Prove that for a hexagonal geometry the co-channel reuse ratio is given by $Q = \sqrt{3N}$; Where $N = i^2 + j^2 + ij$. A cellular service provider decides to use a digital TDMA scheme which can tolerate a Signal-to-Interference Ratio of 15 dB in the worst case. Find the optimal value of N for

- Omni directional antennas
- 120° Sectoring
- 60° Sectoring

[Use path loss exponent of 4 and consider trunking efficiency]

- **Step 1: convert SIR to a linear ratio.** The formula is in power ratio, not dB.

$$\frac{S}{I} = 10^{15/10} = 10^{1.5} = 31.6228$$

- **Step 2: apply** $N = \frac{1}{3} (i_0 \cdot S/I)^{2/n}$ for each antenna configuration. Equivalently find Q first, then $N = Q^2/3$.

Configuration	i_0	$Q = (i_0 S/I)^{1/n}$	$N = Q^2/3$	Round up to allowed N
Omni-directional	6	$(6 \times 31.6228)^{1/4} = (189.74)^{0.25} = 3.7114$	$3.7114^2/3 = 4.5915$	$N = 7$
120° sectoring	2	$(2 \times 31.6228)^{1/4} = (63.246)^{0.25} = 2.8201$	$2.8201^2/3 = 2.6509$	$N = 3$
60° sectoring	1	$(1 \times 31.6228)^{1/4} = 2.3714$	$2.3714^2/3 = 1.8745$	$N = 3$

- **Step 3: why round up, not to the nearest.** N must be one of 1, 3, 4, 7, 9, 12, ..., and a **smaller N gives a worse SIR**. Rounding down would break the 15 dB requirement. **So always take the next allowed value above the computed one.**
 - omni: 4.5915 → next allowed is 7 (4 is allowed but $4 < 4.5915$),
 - 120°: 2.6509 → 3,
 - 60°: 1.8745 → 3.
- **Step 4: comment on the results (the marks for “considering trunking efficiency”).**
 - Sectoring cuts the required cluster size from **7 to 3**, so each cell keeps $7/3 = 2.33$ times as many channels.
 - But 60° sectoring gives the **same** $N = 3$ as 120° here, while splitting each cell’s channels into **6** pools instead of 3.

- A pool of 6 smaller groups carries **less** traffic at the same GoS than 3 larger ones. **So 60° buys no extra capacity and loses more trunking efficiency.**
- **Conclusion: 120° sectoring with $N = 3$ is the best choice.**

1.2 SIR = 15 dB worst case, $n = 3$ (79 Bh)

QUESTION AS PRINTED 2079 Bhadra · Q2 · [3+5]

Define Grade of Service (GoS) and explain how can it be measured in a 'blocked call cleared' type of trunking system. A cellular service provider decides to use a digital TDMA scheme that can tolerate a signal-to-interference ratio of 15 dB in the worst case. The mobile radio channel provided a propagation path loss exponent of $n = 3$. Find the optimal value of N for (a) Omni-directional antennas, (b) 1200 sectoring, and (c) 60° sectoring. Comment on your results.

"1200 sectoring" and "(c)" are the paper's own typos for 120° and (c).

- Same $S/I = 31.6228$, but now the exponent is $2/n = 2/3$.

Configuration	i_0	$Q = (i_0 S/I)^{1/3}$	$N = Q^2/3$	Round up
Omni	6	$(189.74)^{1/3} = 5.7462$	$5.7462^2/3 = 11.0064$	$N = 12$
120°	2	$(63.246)^{1/3} = 3.9842$	$3.9842^2/3 = 5.2913$	$N = 7$
60°	1	$(31.6228)^{1/3} = 3.1623$	$3.1623^2/3 = 3.3333$	$N = 4$

- **Comment.** A smaller path loss exponent means interference decays more slowly, so co-channel cells must sit further apart and N rises sharply ($7 \rightarrow 12$ for omni).
- **Sectoring is therefore even more valuable in a low- n environment:** it takes N from 12 down to 4, a $3\times$ capacity gain.

1.3 The AMPS case, SIR = 18 dB, $n = 4$ (standard textbook check)

PROBLEM not a PYQ · the standard AMPS check verified/added

An AMPS operator requires a worst-case signal-to-interference ratio of 18 dB with a path loss exponent $n = 4$. Find the smallest allowable cluster size N for omni-directional antennas.

- $S/I = 10^{1.8} = 63.0957$.
- Omni: $Q = (6 \times 63.0957)^{1/4} = (378.57)^{0.25} = 4.4110$, so $N = 4.4110^2/3 = 6.4857 \rightarrow \boxed{N = 7}$.
- **This is exactly why every early AMPS system used a 7-cell cluster.**

1.4 Exact Vs approximate SIR for $N = 7$, $n = 4$

PROBLEM not a PYQ · the follow-up every examiner likes verified/added

For a 7-cell cluster with $n = 4$, compute the worst-case S/I twice: first with the approximation that all six first-tier interferers lie at the same distance D , and then exactly, with a mobile at the cell edge seeing interferers at $D - R$, D , D , D , D and $D + R$. Compare the two, in dB.

- **Approximate** (all six interferers taken at distance D):

$$\frac{S}{I} = \frac{(\sqrt{3 \times 7})^4}{6} = \frac{4.5826^4}{6} = \frac{441.0}{6} = 73.50 = 18.66 \text{ dB}$$

- **Exact** (mobile at the cell edge, true distances $D - R$, D , D , D , D , $D + R$ with $Q = D/R = 4.5826$):

$$\frac{S}{I} = \frac{1}{2(Q-1)^{-4} + 2(Q+1)^{-4} + 2Q^{-4}} = 53.376 = 17.27 \text{ dB}$$

- **The approximation is optimistic by about 1.4 dB.** Both fall short of the 18 dB AMPS target, which is the standard justification for sectoring.

Problem 2 Spectrum, duplex channels and channels per cell

PIN 1/22 70 Bh 3+4+3

QUESTION AS PRINTED 2070 Bhadra · Q2 · [3+4+3]

Explain the difference between co-channel and adjacent channel interference. Prove that the co-channel reuse ratio is given by $Q = \sqrt{3N}$, where $N = i^2 + ij + j^2$ is the cluster size. If 20 MHz of total spectrum is allocated for a duplex (i.e. bidirectional) wireless cellular system and each simplex (i.e. one-way) channel has 25 KHz of bandwidth, find

- The number of duplex channels, and

- b) The total number of channels per cell, assuming a cluster size of $N = 4$.

The $Q = \sqrt{3N}$ proof is Problem 1 above; only the spectrum arithmetic is new here.

- **Step 1: what one duplex channel actually costs.** A duplex (two-way) channel needs **two** simplex channels, one for each direction.

$$B_{\text{duplex}} = 2 \times 25 \text{ kHz} = 50 \text{ kHz}$$

Missing this factor of 2 is the single most common error in this question.

- **Step 2: total duplex channels.**

$$N_{\text{total}} = \frac{20 \times 10^6}{50 \times 10^3} = \boxed{400 \text{ duplex channels}}$$

- **Step 3: channels per cell for $N = 4$.**

$$k = \frac{S}{N} = \frac{400}{4} = \boxed{100 \text{ channels per cell}}$$

- **Step 4: the other cluster sizes, for comparison.**

- $N = 7$: $400/7 = 57.14 \rightarrow 57$ channels per cell,
- $N = 12$: $400/12 = 33.33 \rightarrow 33$ channels per cell.

- **Always round down here:** a fractional channel cannot be allocated.

Problem 3 Minimum co-channel distance from the cell area

PIN 1/22 74 Bh 4

QUESTION AS PRINTED 2074 Bhadra · Q2 (b) · [4]

For a seven cell reuse pattern, find the minimum distance between centers of co-channel cells. Area of each cell is uniform and is equal to 23 square km.

- **Step 1: get R from the area.** The cell is a regular hexagon of “radius” R (centre to vertex), so

$$A = \frac{3\sqrt{3}}{2} R^2 = 2.598 R^2$$

$$R = \sqrt{\frac{A}{2.598}} = \sqrt{\frac{23}{2.598}} = \sqrt{8.8530} = 2.9754 \text{ km}$$

- **Step 2: apply the reuse ratio.** For $N = 7$,

$$Q = \sqrt{3N} = \sqrt{21} = 4.5826$$

- **Step 3: minimum co-channel distance.**

$$D = Q \cdot R = 4.5826 \times 2.9754 = \boxed{13.635 \text{ km}}$$

- **Sanity check.** $D/R = 13.635/2.9754 = 4.58 \checkmark$ matches $\sqrt{21}$.
- **If the question had given the area as a circle instead,** use $A = \pi R^2$ and $R = \sqrt{23/\pi} = 2.7052 \text{ km}$, giving $D = 12.397 \text{ km}$. Read the wording carefully: “uniform cell” in this paper means the hexagon.

Problem 4 Users supported and market penetration

HOT 3/22 80 Ba, 81 Ba 5, 79 Ch

QUESTION AS PRINTED 2080 Baishakh · Q2 (a) · [5] · 2079 Chaitra · Q2 (a) · [5]

An urban area has a population of two million residents. System A has 394 cells with 19 channels each. Find the number of users that can be supported at 2% blocking if each user average two calls per hour at average call duration of three minutes. Assuming that trunk systems are operated at maximum capacity, compute the percentage market penetration of cellular provider.

Both papers print this word for word (79 Ch reads “averages two calls”), and 79 Ch attaches the Erlang B table with the $N = 19$ row: 11.23 at 1%, 12.33 at 2%.

- **Step 1: traffic intensity per user.** $A_u = \lambda H$, and both terms must be in the *same* time unit.

$$\lambda = 2 \text{ calls/hour}, \quad H = 3 \text{ min} = \frac{3}{60} \text{ hour}$$

$$A_u = 2 \times \frac{3}{60} = \boxed{0.1 \text{ Erlang per user}}$$

- **Step 2: carried traffic per cell, from the Erlang B table.** With $C = 19$ channels and $\text{GoS} = 0.02$,

$$A = 12 \text{ Erlang}$$

(The exact Erlang B value is 12.333 Erlang; the standard table rounds to 12, which is what the marking scheme expects.)

- **Step 3: users per cell.**

$$U = \frac{A}{A_u} = \frac{12}{0.1} = 120 \text{ users per cell}$$

- **Step 4: total subscribers.**

$$U_{\text{total}} = 120 \times 394 = \boxed{47\,280 \text{ subscribers}}$$

- **Step 5: market penetration.**

$$\frac{47\,280}{2\,000\,000} = 0.02364 = \boxed{2.364 \%}$$

- **Note the assumption stated in the question:** “trunked systems operate at maximum capacity”. That is what allows the full 12 Erlang per cell to be used.
- Using the exact 12.333 Erlang instead gives 123.33 users/cell, 48 592 subscribers and 2.430 % penetration. Either is acceptable if the table value used is stated.

Problem 5 Trunking with 1 and 5 channels, then doubled load

PIN 1/22 77 Ch 2+2+2+3

QUESTION AS PRINTED 2077 Chaitra · Q2 · [2+2+2+3]

A mobile radio system where each user averages three calls per hour and each call lasting an average of 5 minutes.

- What is traffic intensity for each user?
- Find the number of users that could use the system with 1% blocking if only one channel is available.
- Find the number of users that could use the system with 1% blocking if five trunked channels are available.
- If the number of users you found in (c) is suddenly doubled, what is the new blocking probability of the five channel trunked mobile radio system? Justify whether the performance is acceptable or not.

No. of Trunks (N)	0.1%	0.2%	0.5%	1%	1.2%	3%	5%	7%
1	0.001	0.002	0.005	0.010	0.012	0.031	0.053	0.075
2	0.046	0.065	0.105	0.153	0.168	0.282	0.381	0.470
3	0.194	0.249	0.349	0.455	0.489	0.715	0.899	1.06
4	0.439	0.535	0.701	0.869	0.922	1.26	1.52	1.75
5	0.762	0.900	1.13	1.36	1.43	1.88	2.22	2.50

Table caption as printed: “Traffic (A) in erlangs for B%”.

- **(a) Traffic intensity per user.**

$$A_u = \lambda H = 3 \times \frac{5}{60} = \boxed{0.25 \text{ Erlang}}$$

- **(b) One channel, GoS = 1 %.** From the printed table, $C = 1$ and $B = 1 \%$ give $A = 0.010$ Erlang.

$$U = \frac{A}{A_u} = \frac{0.010}{0.25} = 0.04 \text{ users}$$

$$\Rightarrow \boxed{0 \text{ users}}$$

Interpretation: a single channel cannot support even one such user at 1 % blocking. A lone user offering 0.25 Erlang to one channel would be blocked about $0.25/(1 + 0.25) = 20 \%$ of the time.

- **(c) Five trunked channels, GoS = 1 %.** From the table, $C = 5$ and $B = 1 \%$ give $A = 1.36$ Erlang.

$$U = \frac{1.36}{0.25} = 5.44 \Rightarrow \boxed{5 \text{ users}}$$

- **This is the whole point of trunking:** 1 channel supports 0 users, but 5 channels support 5. Capacity grows faster than the channel count.
- **(d) Users doubled to 10.**

$$A = U A_u = 10 \times 0.25 = 2.5 \text{ Erlang}$$

Apply the Erlang B formula with $C = 5$, $A = 2.5$:

$$\text{Pr}[\text{blocking}] = \frac{A^C / C!}{\sum_{k=0}^C A^k / k!}$$

k	0	1	2	3	4	5
$A^k/k!$	1	2.5	3.125	2.6042	1.6276	0.8138

$$\sum_{k=0}^5 \frac{A^k}{k!} = 11.6706, \quad \text{Pr[blocking]} = \frac{0.8138}{11.6706} = 0.0697$$

$$\text{GoS} = 6.97 \%$$

- **Justification.** The design target was 1 %. Doubling the users has pushed blocking to **almost 7 %, seven times worse**. **This is not acceptable:** roughly 1 call in 14 fails in the busy hour.
- **Note how non-linear this is:** doubling the load multiplied the blocking by 7, not by 2. That non-linearity is the reason cellular systems are dimensioned on the busy hour.

Problem 6 Full system design from area and spectrum

PIN 1/22 72 Ma 12 · the longest numerical in the chapter

QUESTION AS PRINTED 2072 Magh · Q2 · [12]

A city with a coverage area of 1500 sq km is covered with a 12-cell system each with a radius of 1.387 km. If the total spectrum allocated is 28.5 MHz with a full duplex channel bandwidth of 25 MHz. Assume a GOS of 0.02 for a blocked calls cleared system, is specified and the offered traffic per user is 0.03 Erlangs and traffic intensity of each cell is 84 Erlang, compute:

- the number of cells in the service area
- the number of channels per cell
- the maximum carrier traffic
- the total number of users that can be served for 2% GOS
- the number of mobiles per unique channel.
- Theoretical maximum number of users that could be served at one time by the system.

The paper prints “25 MHz” for the channel bandwidth: a typo for 25 kHz, since 28.5 MHz would otherwise buy one channel. Solve with 25 kHz and say so.

- **(a) Number of cells in the service area.**

$$A_{\text{cell}} = 2.598 R^2 = 2.598 \times (1.387)^2 = 2.598 \times 1.9238 = 4.998 \text{ km}^2$$

$$n_{\text{cells}} = \frac{1500}{4.998} = 300.12 \Rightarrow \boxed{300 \text{ cells}}$$

- **(b) Channels per cell.**

$$N_{\text{total}} = \frac{28.5 \times 10^6}{25 \times 10^3} = 1140 \text{ duplex channels}$$

The question says “full duplex channel bandwidth 25 kHz”, so the $\times 2$ of Problem 2 does *not* apply here. Read the wording.

$$k = \frac{1140}{12} = \boxed{95 \text{ channels per cell}}$$

- **(c) Maximum carried traffic per cell.** The question supplies this as **84 Erlang**. Cross-check from Erlang B: $C = 95$ at $\text{GoS} = 0.02$ gives $A = 83.13$ Erlang, so the given 84 is consistent.

$$A_{\text{cell}} = \boxed{84 \text{ Erlang}}$$

System total = $84 \times 300 = 25\,200$ Erlang.

- **(d) Total users served at 2 % GoS.**

$$U_{\text{cell}} = \frac{A_{\text{cell}}}{A_u} = \frac{84}{0.03} = 2800 \text{ users per cell}$$

$$U_{\text{total}} = 2800 \times 300 = \boxed{840\,000 \text{ users}}$$

- **(e) Mobiles per unique channel.**

$$\frac{U_{\text{cell}}}{k} = \frac{2800}{95} = \boxed{29.47 \text{ mobiles per channel}}$$

le roughly 29 subscribers share each radio channel, which is exactly the statistical multiplexing gain trunking delivers.

- **(f) Theoretical maximum users served at one time.** Every channel in every cell busy simultaneously:

$$95 \times 300 = \boxed{28\,500 \text{ simultaneous calls}}$$

- **Sanity check on (d) against (f):** 840 000 subscribers share 28 500 simultaneous circuits, a ratio of 29.5:1, matching (e). ✓

Problem 7 Area, channels and simultaneous calls

PIN 1/22 82 Bh 4

QUESTION AS PRINTED 2082 Bhadra · Q2 (b) · [4]

The cellular system has 32 cells, each cell has a 1.6 km radius and the system reuse factor is 7. The system is to support 336 traffic channels in total. Determine the total geographical area covered, the number of traffic channels per cell and the total number of simultaneous calls supported by this system.

- **Step 1: area of one cell.**

$$A_{\text{cell}} = 2.598 \times (1.6)^2 = 2.598 \times 2.56 = 6.6509 \text{ km}^2$$

- **Step 2: total geographical area.**

$$A_{\text{total}} = 32 \times 6.6509 = \boxed{212.83 \text{ km}^2}$$

- **Step 3: channels per cell.**

$$k = \frac{S}{N} = \frac{336}{7} = \boxed{48 \text{ channels per cell}}$$

- **Step 4: simultaneous calls.** Every cell can run all 48 of its channels at once, and there are 32 cells:

$$48 \times 32 = \boxed{1536 \text{ simultaneous calls}}$$

- **Watch the trap:** the 336 channels are the **system** total, reused by every cluster. The system supports 1536 calls, not 336, because the cluster repeats $32/7 = 4.57$ times.

Problem 8 Trunking loss from omni to 60° sectoring

PIN 1/22 81 Ba 2+2+2 · the one question where sectoring is shown to cost capacity

QUESTION AS PRINTED 2081 Baishakh · Q2 (a) · [2+2+2]

For a cluster $N = 7$ system with a blocking probability of 1% and an average call length of two minutes, find the traffic capacity loss for a blocked calls cleared system due to trunking for 57 channels when going from omnidirectional antennas to 60° sectorized antennas. The average request rate per user is 1 call per hour.

GOS / Channel	0.5%	1%	2%
1	.005	.0101	.0204
9	3.333	3.783	4.345
19	10.33	11.23	12.33
54	39.47	41.51	44.00
57	42.1	44.2	46.8

- **Step 1: traffic per user.**

$$A_u = \lambda H = 1 \times \frac{2}{60} = 0.0333 \text{ Erlang}$$

- **Step 2: omni-directional case.** All 57 channels sit in **one** trunked pool. From the table, $C = 57$ at 1 % gives

$$A_{\text{omni}} = 44.2 \text{ Erlang}$$

$$U_{\text{omni}} = \frac{44.2}{0.0333} = 1326 \text{ users per cell}$$

- **Step 3: 60° sectorized case.** The cell is split into **6 sectors**, and each sector gets its own channel group:

$$\frac{57}{6} = 9.5 \Rightarrow 9 \text{ channels per sector}$$

Round down: a channel cannot be split between sectors. From the table, $C = 9$ at 1 % gives $A = 3.783$ Erlang per sector, so

$$A_{\text{sectorized}} = 6 \times 3.783 = 22.698 \text{ Erlang}$$

$$U_{\text{sectorized}} = \frac{22.698}{0.0333} = 681 \text{ users per cell}$$

– **Step 4: the loss.**

$$\Delta A = 44.2 - 22.698 = \boxed{21.50 \text{ Erlang lost}}$$

$$\frac{21.50}{44.2} \times 100 = \boxed{48.65 \% \text{ capacity loss}}$$

– **Why this happens (state this, it is where the marks are).**

- Erlang B is **strongly non-linear** in the channel count. One pool of 57 channels carries 44.2 Erlang, but six pools of 9 carry only $6 \times 3.783 = 22.7$ Erlang.
- That is the **loss of trunking efficiency**: a large pool absorbs random peaks far better than several small ones.
- Three channels are also lost outright to the rounding ($57 \rightarrow 6 \times 9 = 54$).
- **The complete engineering picture**: sectoring **improves SIR**, which lets N be reduced, which gives *more* channels per cell. That gain must beat the **48.65 % trunking loss** computed here, otherwise sectoring is a net loss. This problem deliberately shows only the cost side.

Problem 9 Calls per cell per hour, and SIR

lecture-note example, same pattern as 79 Bh and 72 Ma

PROBLEM not a PYQ · lecture-note example verified/added

A cellular system has 416 radio channels available, of which 21 are control channels. The average channel holding time is 3 minutes and the blocking probability during the busy hour is 2%. The frequency reuse factor is 9. Find (i) the number of calls per cell per hour, and (ii) the signal to co-channel interference ratio in dB (take $n = 4$).

– **Step 1: voice channels.**

$$416 - 21 = 395 \text{ voice channels}$$

– **Step 2: voice channels per cell.**

$$\frac{395}{9} = 43.89 \Rightarrow 44 \text{ channels per cell (taken as 44 for the table lookup)}$$

– **Step 3: carried traffic.** From the Erlang B table, $C = 44$ at GoS = 0.02 gives

$$A = 34.683 \text{ Erlang}$$

– **Step 4: convert Erlang to calls.** 1 Erlang = one continuously occupied channel. With a 3-minute holding time, one Erlang carries $1/3$ call per minute.

$$\text{calls per cell per minute} = \frac{34.683}{3} = 11.561$$

$$\text{calls per cell per hour} = 11.561 \times 60 = \boxed{694 \text{ calls/cell/hour}}$$

– **Step 5: SIR for $N = 9, n = 4$.**

$$Q = \sqrt{3 \times 9} = \sqrt{27} = 5.1962$$

$$\frac{S}{I} = \frac{Q^4}{6} = \frac{729}{6} = 121.5$$

$$\frac{S}{I} [\text{dB}] = 10 \log_{10}(121.5) = \boxed{20.85 \text{ dB}}$$

- **Comment:** 20.85 dB comfortably exceeds the 18 dB AMPS requirement, so $N = 9$ is a conservative, high-quality plan. $N = 7$ would give 18.66 dB, barely adequate.

Problem 10 Cell splitting: the transmit power reduction

HOT 3/22 73 Bh, 75 Bh, 81 Bh · usually a 1 or 2 mark part inside the sectoring Vs splitting question

QUESTION AS PRINTED 2081 Bhadra · Q3 · [4] · 2075 Bhadra · Q3 · [5]

A telephone network company needs to expand its capacity based on demand on a city. A group of engineers was selected to find the solution. Among the solution sectoring and cell splitting were major technique for expansion purpose. Being a cellular planning engineer, which option do you think is best and why?

QUESTION AS PRINTED 2073 Bhadra · Q2 (a) · [5]

Explain how cell splitting and sectoring improve coverage and capacity in cellular system?

No paper prints numbers for this. The transmit-power result below is the only computable part, and it is what turns a prose answer into a full-mark one.

verified/added

- **Requirement.** The new microcell of radius $R/2$ must deliver the **same received power at its own boundary** as the old cell did at its boundary. Only then does the reuse plan carry over unchanged.
- **Step 1: write both boundary powers.**

$$P_r[\text{old boundary}] \propto P_{t1} R^{-n}, \quad P_r[\text{new boundary}] \propto P_{t2} \left(\frac{R}{2}\right)^{-n}$$

- **Step 2: set them equal.**

$$P_{t1} R^{-n} = P_{t2} \left(\frac{R}{2}\right)^{-n} \Rightarrow \frac{P_{t2}}{P_{t1}} = \left(\frac{1}{2}\right)^n$$

- **Step 3: put $n = 4$.**

$$\frac{P_{t2}}{P_{t1}} = \frac{1}{16} = 0.0625$$

$$10 \log_{10}(0.0625) = \boxed{-12.04 \text{ dB, ie reduce power by 12 dB}}$$

- **Capacity effect.** Halving R needs about **4 times as many cells** ($A \propto R^2$), so the number of clusters, and therefore the capacity, rises about **4 times** for the same spectrum.

Erlang B reference table

Values recomputed from $\Pr[\text{blocking}] = \frac{A^C / C!}{\sum_{k=0}^C A^k / k!}$ and cross-checked against the tables printed on the 77 Ch and 81 Ba papers. **Both printed tables agree with this one** once their differing row labels ("Trunks N " Vs "GOS / Channel") are read correctly.

Channels C	A at 0.5 %	A at 1 %	A at 2 %	A at 5 %
1	0.005	0.010	0.020	0.053
2	0.105	0.153	0.223	0.381
3	0.349	0.455	0.602	0.899
4	0.701	0.869	1.092	1.525
5	1.132	1.361	1.657	2.219
9	3.333	3.783	4.345	5.370
10	3.961	4.461	5.084	6.216
19	10.331	11.230	12.333	14.315
20	11.092	12.031	13.182	15.249
44	30.797	32.543	34.682	38.557
54	39.474	41.505	43.997	48.536
57	42.109	44.222	46.816	51.548
95	76.325	79.368	83.133	90.123

How to use it in the exam: you are always given either C and the GoS (look up A , then divide by A_u for users), or C and A (look across the row for the GoS). Never try to sum the series by hand unless $C \leq 5$.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
BS	base station	BTS	base transceiver station
MS	mobile station	MSC	mobile switching centre
BSC	base station controller	MSK	minimum shift keying
CCI	co-channel interference	ACI	adjacent channel interference
SIR, S/I	signal to interference ratio	SNR, S/N	signal to noise ratio
GoS	grade of service	MAHO	mobile assisted handoff
FCA	fixed channel assignment	DCA	dynamic channel assignment
LOS	line of sight	NLOS	non line of sight
FDMA	frequency division multiple access	TDMA	time division multiple access
CDMA	code division multiple access	OFDM	orthogonal frequency division multiplexing
FDD	frequency division duplex	TDD	time division duplex
GSM	global system for mobile communications	AMPS	advanced mobile phone system
PCS	personal communication system	DAS	distributed antenna system
BCCH	broadcast control channel	EIRP	effective isotropic radiated power
ERP	effective radiated power	RF	radio frequency
Tx, TX	transmitter	Rx, RX	receiver
ISI	intersymbol interference	PDP	power delay profile
DFE	decision feedback equalizer	MLSE	maximum likelihood sequence estimation
LMS	least mean square	RLS	recursive least squares
MSE	mean square error	MMSE	minimum mean square error
ZF	zero forcing	MRC	maximal ratio combining
EGC	equal gain combining	BER	bit error rate
PCM	pulse code modulation	DPCM	differential PCM
ADPCM	adaptive differential PCM	LPC	linear predictive coder
RPE-LTP	regular pulse excited, long term prediction	CELP	code excited linear prediction
RELp	residual excited linear prediction	MPE	multipulse excited
VSELP	vector sum excited linear prediction	ATC	adaptive transform coding
SBC	sub-band coding	DCT	discrete cosine transform
FEC	forward error correction	ARQ	automatic repeat request
BCH	Bose–Chaudhuri–Hocquenghem code	RS	Reed–Solomon code
TCM	trellis coded modulation	PN	pseudo-noise

Short	Means	Short	Means
MIMO	multiple input multiple output	OFDMA	orthogonal FDM access
QAM	quadrature amplitude modulation	PSK	phase shift keying
GMSK	Gaussian minimum shift keying	IMT	international mobile telecommunications
ASK	amplitude shift keying	FSK	frequency shift keying
BPSK	binary phase shift keying	QPSK	quadrature phase shift keying
OQPSK	offset (staggered) QPSK	DPSK,	differential (binary) PSK
		DBPSK	
CPFSK	continuous phase FSK	MFSK,	M-ary FSK, M-ary PSK
		MPSK	
DSSS	direct sequence spread spectrum	FHSS	frequency hopped spread spectrum
SS, SSM	spread spectrum (modulation)	PG	processing gain
PAPR	peak-to-average power ratio	CP	cyclic prefix
FFT, IFFT	(inverse) fast Fourier transform	ICI	inter-carrier interference
PSD	power spectral density	CFO, SFO	carrier, sampling frequency offset
NRZ, RZ	non-return, return to zero	AMI	alternate mark inversion
AM, FM, PM	amplitude, frequency, phase modulation	DSB, SSB, VSB	double, single, vestigial sideband
S/P, P/S	serial-parallel converter	D/A, A/D	digital-analog converter
LPF, BPF	low pass, band pass filter	ISM	industrial, scientific, medical band
SC-FDMA	single carrier FDMA	<i>BT</i>	GMSK filter bandwidth $\times$ symbol time
PSTN	public switched telephone network	VoIP	voice over IP
FVC, RVC	forward, reverse voice channel	FCC, RCC	forward, reverse control channel
SSMA	spread spectrum multiple access	SDMA	space division multiple access
FHMA	frequency hopped multiple access	DSMA	direct sequence multiple access (= CDMA)
FCDMA	hybrid FDMA/CDMA	DS/FHMA	hybrid direct sequence / frequency hopped
TCDMA	time division CDMA	TDFH	time division frequency hopping
IM	intermodulation (product)	MU-MIMO	multi-user MIMO
WLAN	wireless local area network	WMAN	wireless metropolitan area network
WPAN	wireless personal area network	AP	access point
BSS, IBSS	basic service set, independent (ad hoc) BSS	ESS	extended service set
CSMA/CA	carrier sense multiple access with collision avoidance	CSMA/CD	... with collision detection
DCF	distributed coordination function	PCF	point coordination function
RTS/CTS	request to send / clear to send	CCK	complementary code keying
WEP, WPA	wired equivalent privacy, WiFi protected access	U-NII	unlicensed national information infrastructure band
IoT	internet of things	mmWave	millimetre wave
eMBB	enhanced mobile broadband	URLLC	ultra-reliable low latency comm.
mMTC	massive machine type comm.	NOMA	non-orthogonal multiple access
D2D	device to device	C-RAN	cloud radio access network
PCN	personal communication network	SMS	short message service
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and

Short	Means	Short	Means
BSS	base station subsystem	NSS	network and switching subsystem
OSS	operation support subsystem	OMC	operation and maintenance centre
HLR	home location register	VLR	visitor location register
AuC	authentication centre	EIR	equipment identity register
CEIR	central EIR	ME	mobile equipment
SIM	subscriber identity module	IMSI	international mobile subscriber identity
IMEI	international mobile equipment identity	TMSI	temporary mobile subscriber identity
Um, Abis, A	the three GSM interfaces	SS7	signalling system no. 7
ARFCN	absolute radio frequency channel number	TN, TS	time slot number, time slot
TCH	traffic channel	TCH/FS, TCH/HS	full rate, half rate speech TCH
CCH	control channel	CCCH	common control channel
DCCH	dedicated control channel	SDCCH	stand-alone dedicated control channel
SACCH	slow associated control channel	FACCH	fast associated control channel
FCCH	frequency correction channel	SCH	synchronisation channel
PCH	paging channel	RACH	random access channel
AGCH	access grant channel	TB, GP, SF	tail bits, guard period, stealing flag
VAD	voice activity detector	DTX	discontinuous transmission
DCS 1800	digital cellular system, the 1800 MHz GSM	CRC	cyclic redundancy check
ESN	electronic serial number	PCB	power control bit
Mcps, Ksps	mega-chips, kilo-symbols per second	PRAT	paging channel rate field
EVRC	enhanced variable rate coder	PDSN	packet data serving node
PCF	packet control function	RNC	radio network controller
(CDMA)			
EPS, EPC	evolved packet system, core	eNodeB	LTE base station
MME	mobility management entity	S-GW, P-GW	serving, packet data network gateway
HSS	home subscriber server	PCRF	policy and charging rules function
IMS, VoLTE	IP multimedia subsystem, voice over LTE	CoMP	coordinated multipoint
ITU-R	ITU radiocommunication sector	WRC	world radiocommunication conference
NTA	Nepal Telecommunications Authority	UGS, rtPS, nrtPS	WiMAX QoS service classes

Standard and system names (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **USDC** US digital cellular (IS-136) · **JDC** Japanese digital cellular (later PDC) · **Bluetooth** IEEE 802.15.1 · **Zigbee** IEEE 802.15.4 · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols · **CDPD** cellular digital packet data · **Mobitex** a dedicated packet data system · **TETRA** terrestrial trunked radio · **DAB** digital audio broadcasting · **DVB-T** digital video broadcasting, terrestrial · **ADSL** asymmetric digital subscriber line · **5G NR** 5G new radio.

Three clashes to watch: **FCC** is the *forward control channel* throughout these notes; the US regulator is always written out as the Federal Communications Commission, and **FCCH** is GSM's frequency correction channel, a different thing again. **BCH** is the Bose–Chaudhuri–Hocquenghem block code in **Ch 6** and GSM's *broadcast channel* in **Ch 8**. **SCH** is GSM's synchronisation channel, never a sub-channel.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams. **Monospace** year tags are BEI (EX 715); the rest are BEX (EX 751).

Tier chips over 22 papers, 2070–2082 BS: **TOP /22** = topic asked in 6 or more papers · **HOT /22** = 3 to 5 papers · **PIN /22** = 1 or 2 papers but worth ≥ 6 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.